

Hallucination in LLMs

Why language models confidently say things that aren't true, and what engineers can do about it.



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Hallucination Has Real Consequences



In 2023, a New York lawyer was sanctioned after submitting a brief containing fabricated case citations generated by ChatGPT. The model invented plausible sounding but nonexistent precedents.



Legal RAG tools from LexisNexis and Thomson Reuters still hallucinate between 17 and 33 percent of the time, even with retrieval grounding, per a 2025 Stanford empirical study.



A 2025 Scientific Reports analysis of three million mobile app reviews found that roughly 1.75 percent of user complaints explicitly described hallucination like errors in everyday AI features.



Hallucination is not a fixable bug. Learning theory research published at arXiv shows it is a provably inevitable property of any general purpose LLM used outside the scope of its training distribution.

THE FLUENCY TRAP

Hallucinated text reads exactly like correct text. The model's writing quality gives no signal that a fact is fabricated. Fluency and truthfulness are entirely orthogonal properties in LLMs.



3 Ways an LLM Hallucinates



INTRINSIC HALLUCINATION

The model generates output that directly contradicts facts in its own training data or in the user's provided context. It knows the answer but produces the wrong one anyway, often because the conflicting fact was underrepresented during pre training. Example: a model states that a historical event happened in 1945 when it occurred in 1953.



EXTRINSIC HALLUCINATION

The model fabricates content that cannot be verified or contradicted by any source it was given. Invented citations, nonexistent API endpoints, and fictional statistics fall here. The model has no fact to contradict because the fact never existed to begin with.



FACTUALITY HALLUCINATION

The model generates statements that are syntactically perfect and contextually plausible but factually wrong against external ground truth. These are the most dangerous in production because they pass basic coherence checks. A confident wrong answer to a medical or legal question is a factuality hallucination.



Training Rewards Confident Guessing

THE INCENTIVE PROBLEM BAKED INTO EVERY BENCHMARK

1 Next token prediction does not encode factual truth

LLMs predict the statistically likely next token, not the factually correct one. A token that sounds right given the sentence pattern scores well even when it is wrong.

2 Accuracy only benchmarks penalize admitting uncertainty

On standard leaderboards, guessing has a 1-in-365 chance of being right. Saying 'I do not know' scores zero. Over thousands of questions, the guesser ranks higher than the honest model.

3 RLHF alignment can amplify fluency over truth

Human raters reward responses that sound confident. A hedged but accurate answer often scores lower than a confident but wrong one, pushing models toward plausible sounding fabrication.



THE SCORECARD DISTORTION

OpenAI's September 2025 paper shows this is systemic: leaderboards that measure accuracy but not calibration actively incentivize hallucination. Fixing evals is as important as fixing models.



The Gaps in Parametric Memory

WHAT THE MODEL DOES NOT KNOW STILL GETS ANSWERED

1 Rare and niche facts are underrepresented in training data

Pre training corpora reflect the web. Obscure events and specialized domains appear far less often, leaving the model with weak signal and high fabrication risk on niche queries.

2 Knowledge has a hard cutoff date

Any event released after training does not exist in parametric memory. Querying post cutoff facts forces the model to extrapolate from outdated patterns, producing confident but stale answers.

3 Training data noise propagates directly into model beliefs

Web scraped corpora contain errors and AI generated text. A model trained on inaccurate claims absorbs those as valid patterns, making some hallucinations a direct replay of corrupted training signal.



THE CONFIDENT EXTRAPOLATION

The model cannot distinguish between what it knows and what it has inferred from patterns. Ask it about a person who became famous after its cutoff and it will construct a plausible but fabricated biography.



TRANSFORMER ARCHITECTURE CONTRIBUTES TO FABRICATION

POSITIONAL BIAS

OVERCONFIDENCE IN GENERATION

THE LOST IN THE MIDDLE EFFECT

Research shows retrieval accuracy drops sharply for facts placed in the middle of long contexts. Keep critical grounding evidence near the start or end of your prompt, not buried in the middle.



RAG: Grounding Beats Memory

CONNECT THE MODEL TO VERIFIABLE EXTERNAL SOURCES

1 Retrieve relevant chunks before the model generates

A dense or hybrid retriever fetches the top-k document segments and injects them into the prompt. The model generates from retrieved evidence, not from parametric pattern matching.

2 Hybrid retrieval outperforms dense retrieval alone

Combining BM25 keyword search with FAISS dense vector retrieval improves recall and factual grounding. Keyword matching catches exact names and dates that vector similarity misses on rare entities.

3 Instruct the model to stay within retrieved context

Directives like 'answer only from the documents below' reduce out of context generation. When retrieved and parametric knowledge conflict, route to the retrieved source unless explicitly overridden.



RAG IS NOT A SILVER BULLET

Legal AI tools using RAG still hallucinate 17 to 33 percent of the time per a 2025 Stanford study. RAG reduces the problem but does not solve it. Pair it with output validation for high stakes tasks.



Mitigation Methods Compared

Method	Effort	Latency	Reduction
RAG Grounding	Medium	Low overhead	35 to 60% errors
Chain-of-Thought	Low	Moderate increase	Prompt sensitive
Fine tuning on facts	High	None after train	Domain specific
Temperature 0.1-0.4	Very low	None	Reduces variance
Guardrails + validation	Medium	Under 200ms	Up to 97% detect
Self-consistency	Low code	3-5x slower	Strong for math



STACK YOUR DEFENSES

No single method eliminates hallucination. RAG plus guardrails plus low temperature is a standard production stack. Add self-consistency sampling only for high stakes outputs where latency permits.



Prompts That Fight Fabrication

WRITE PROMPTS THAT CONSTRAIN, NOT INVITE

1 Chain-of-Thought Cuts Prompt Sensitive Errors

Instructing the model to reason step by step forces intermediate claims to surface. GPT-4 and LLaMA 2 research shows CoT cuts errors significantly on multi step queries.

2 Explicit uncertainty instructions change model behavior

Adding 'if uncertain, say so explicitly' to your system prompt makes hedging acceptable. Pairing with 'do not guess' measurably shifts the model toward refusal over fabrication.

3 Keep temperature between 0.1 and 0.4 for factual tasks

High temperature inflates hallucination rate on factual queries. Set it to 0.1 for anything involving citations, dates, statistics, or domain specific facts.



THE REPETITION TRICK

Restating the key constraint at both the beginning and end of a prompt reduces mid generation drift. Phrase the end constraint differently to avoid verbatim repetition triggering positional attention bias.



DATA QUALITY IS A HALLUCINATION CONTROL LEVER

Errors in fine tuning data propagate directly into model behavior. A single mislabeled fact repeated across examples can become a confident false belief. Data cleaning is not optional.

AI generated content is a significant fraction of publicly available web text. RAG pipelines that pull from the open web without filtering are retrieval augmented hallucination systems.



Guardrails: Catch It Before It Ships

BUILD A VERIFICATION LAYER AROUND EVERY LLM CALL

1 Hybrid RAG Plus Validation Reaches 97% Detection

Combining AWS Bedrock contextual grounding with NVIDIA NeMo guardrails achieves state of the art detection under 200ms latency. Run this layer on every response before it reaches the user.

2 Self-consistency sampling catches logical hallucinations

Generate three to five responses at low temperature and compare. Where answers diverge the model is uncertain. Flag those responses for human review rather than auto serving.

3 Flag low confidence outputs for human review

Token probability scores from logit outputs measure uncertainty. Set a confidence threshold and route any response below it to a human review queue before serving.



AUTOGEN STYLE MULTI AGENT VALIDATION

Multi agent systems where one model critiques another's output reduce critical errors by 82 percent in high stakes domains. A critic agent is worth more than a better base model alone.



Pick Your Mitigation Stack in 4 Steps

RUN THESE CHECKS BEFORE DEPLOYING ANY LLM FEATURE

1

Classify Task Risk

Is this high stakes (medical, legal, financial) or low stakes (creative)? High stakes tasks require guardrails plus human review. Low stakes tasks can use higher temperature.

2

Decide on Grounding

Does the task require facts beyond training or post cutoff? If yes, RAG is mandatory. Build the retrieval pipeline before touching prompt engineering.

3

Set Temperature First

Temperature is the fastest lever. Drop to 0.1 to 0.2 for factual queries before any prompt tuning. Low temperature alone cuts hallucination more than elaborate prompts at 0.9.

4

Add Validation Last

Wire NeMo Guardrails, Bedrock contextual grounding, or a critic agent before going live. Validation is not a post incident fix. It is a precondition for shipping.



HALLUCINATION TRAPS IN PRODUCTION

🔴 Running factual tasks at high temperature: temperature above 0.5 inflates hallucination rate on names, dates, and statistics. Lock factual endpoints to 0.1 to 0.2 first.

 Burying grounding context in the middle of long prompts: attention deprioritizes the middle of long contexts. Put critical retrieved facts at the start of the prompt, not the middle.

- 🔍 Skipping output validation before launch: guardrails are a precondition for shipping, not a post incident retrofit. Wire NeMo or Bedrock contextual grounding before go live.

🧠 Confusing fluency with accuracy: hallucinated text reads identically to correct text. Teams without automated factual checks routinely approve polished but wrong AI outputs.



A Production Reality Check

“

An LLM that cannot say it does not know is not a reasoning system. It is an autocomplete engine with a confidence problem. Build for calibrated uncertainty, not for the appearance of certainty.

— Rajesh Mane, Senior AI Engineer



Key Takeaways



Hallucination is structural

LLMs predict plausible tokens, not true ones. Better data and calibration-aware evals are the only paths to systemic reduction.



RAG grounds outputs

RAG anchors answers in retrieved documents. Use hybrid retrieval combining dense vectors and BM25 keyword search for best coverage.



Guardrails are mandatory

NeMo Guardrails and Bedrock contextual grounding achieve high detection rates under 200ms. Wire validation in before launch, not after.



Prompts and temp matter

Chain-of-thought prompting and temperature 0.1 to 0.2 cut prompt sensitive hallucinations at zero infrastructure cost.



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